

Professional-Skepticism Engine

Per-Agent Input / Output — one case traced end-to-end, with the final report

This document traces a single audit case — **CASE-0007 / TXN-0876**, a \$58,000 payment to “Mehta Family Holdings” (Consulting) — through every agent, showing the concrete input each agent receives and the structured output it returns, ending with the final case report and audit entry.

Pipeline at a glance

Agent	Consumes	Produces
Supervisor · Planner	case	plan · routing · escalation
Evidence agent	case	evidence{} + citations
Challenger	case + evidence	challenger_view (concern)
Defender	case + evidence	defender_view (legitimacy)
Adjudicator	evidence + both views	adjudication{risk·conf·disposition}
Verifier · QA	adjudication + evidence	verification{passed·issues}
Confidence gate	adjudication + verification	routing{auto·review·manual}
Human review	full bundle + transcript	review{action·comment·signature}
Report / Audit	everything	case file + audit_history[]

The case (input to the pipeline)

```
case = {
  "case_id": "CASE-0007", "txn_id": "TXN-0876",
  "date": "2026-05-20", "account": "Consulting",
  "vendor": "Mehta Family Holdings",
  "description": "Consulting - related party (director J. Mehta)",
  "currency": "USD", "amount": 58000, "amount_usd": 58000,
  "posted_by": "rkumar",
  "flags": ["round-dollar amount", "statistical outlier (Consulting)",
    "related-party indicator"]
}
```

Agent-by-agent input / output

1 · Supervisor · Planner

plans & routes the investigation

INPUT

case # the flagged transaction above

OUTPUT

```
plan = {
  "steps": ["gather_evidence", "run_debate", "adjudicate", "verify"],
  "specialists": ["evidence", "challenger", "defender", "adjudicator", "verifier"],
  "max_debate_rounds": 2,
  "materiality_usd": 25000,
  "escalation": "partner_if_material",
  "route_hint": "material -> expect manual / partner"
}
```

2 · Evidence agent

RAG + history + external data

INPUT

case

OUTPUT

```
evidence = {
  "policy": [
    {"id": "RP-01 §2", "v": "2.1", "text": "Related-party txns require independent review (ISA 550)."},
    {"id": "AP-02 §1", "v": "3.0", "text": ">$5k needs manager approval; >$25k needs partner approval."}],
  "history": {"vendor_prior_12m": 0, "account_mean_usd": 15126, "z_score": 4.1},
  "external": {"fx": "n/a (USD)",
    "vendor_master": "Mehta Family Holdings – NOT in approved vendor master"},
  "citations": ["RP-01 §2", "AP-02 §1", "history:Consulting", "vendormaster"],
  "api_metadata": [{"source": "vendor_master", "status": 200, "cached": false}]
}
```

3 · Challenger (skeptic)

argues the worst case

INPUT

case + evidence

OUTPUT

```
challenger_view = {
  "position": "concern", "severity": "high",
  "claims": [
    "Vendor implies related party to director J. Mehta (RP-01 §2).",
    "$58,000 is 4.1σ above Consulting mean AND an exact round number.",
    "Vendor absent from approved master; no 12-month history (ghost vendor).",
    "$58,000 exceeds the $25,000 partner-approval threshold (AP-02 §1)."],
  "red_flags": ["related_party", "round_dollar", "outlier", "ghost_vendor", "approval_breach"]
}
```

4 · Defender

argues legitimacy (independently)

INPUT

case + evidence # does NOT see the Challenger's reasoning

OUTPUT

```
defender_view = {
  "position": "legitimate_possible", "confidence_in_legitimacy": 0.25,
  "arguments": [
    "Large one-off strategy retainers occur in Consulting; can be round by contract.",
    "Related-party is permitted IF independently approved and disclosed.",
    "A new vendor naturally has no 12-month history."],
  "conditions": ["signed contract", "independent approval", "related-party disclosure on file"]
}
```

5 · Adjudicator · Lead auditor

weighs both, decides

INPUT

evidence + challenger_view + defender_view

OUTPUT

```
adjudication = {
  "risk_level": "High", "confidence": 0.86,
  "disposition": "investigate_before_reliance",
  "rationale": "Related-party + 4.1σ round-number outlier + ghost vendor + breach of the "
    "AP-02 partner-approval threshold outweighs the benign-retainer hypothesis, "
    "which has NO approval/disclosure evidence on file.",
  "citations": ["RP-01 §2", "AP-02 §1", "history:Consulting z=4.1", "vendormaster:absent"]
}
```

6 · Verifier · QA

checks grounding, catches hallucination

INPUT

adjudication + evidence

OUTPUT

```
verification = {
  "passed": true, "checked_claims": 4, "grounded": 4, "ungrounded": 0,
  "notes": "Every adjudicator claim maps to a retrieved evidence id; no unsupported assertions.",
  "action": "proceed"
}
# If a claim were unsupported -> {"passed": false, "action": "revise"} -> loop to Adjudicator
```

7 · Confidence gate

routes by confidence + materiality

INPUT

adjudication + verification

OUTPUT

```
routing = {
  "decision": "manual",
  "reason": "amount $58,000 >= materiality $25,000 AND risk=High -> partner escalation"
}
```

INPUT

full bundle: case + evidence + debate_transcript + adjudication + verification

OUTPUT

```
review = {
  "reviewer_id": "a.sharma (Partner)", "action": "reject_pending_evidence",
  "comment": "Hold payment. Obtain signed contract + independent approval + related-party "
    "disclosure before any reliance.",
  "signature": "sha256:9f2c4b...e71", "timestamp": "2026-06-25T11:42:08Z"
}
```

Final output sample — case report

This is the artifact the Report agent produces (rendered here as it would appear in the dashboard / workpaper).

HIGH	CASE-0007 · TXN-0876	Mehta Family Holdings	Consulting	\$58,000
Date / posted by	2026-05-20 · rkumar · USD 58,000.00			
Flags	round-dollar amount · statistical outlier (Consulting, z=4.1) · related-party indicator			
Evidence	Vendor NOT in approved master · 0 prior 12-mo activity · account mean \$15,126			
Debate — concern	related party (RP-01 §2) · 4.1σ round outlier · ghost vendor · breach of AP-02 \$25k partner approval			
Debate — defence	possible one-off retainer, but no contract / approval / disclosure on file (conf 0.25)			
Verdict	High · confidence 0.86 · disposition: investigate before reliance			
Verifier	passed — 4/4 claims grounded			
Citations	RP-01 §2 · AP-02 §1 · history:Consulting z=4.1 · vendormaster:absent			
Routing	manual — partner escalation (≥ materiality \$25,000)			
Decision	a.sharma (Partner): reject_pending_evidence — hold payment; obtain contract + approval + disclosure			

Final output — machine-readable case file

```
{
  "case_id": "CASE-0007", "txn_id": "TXN-0876", "vendor": "Mehta Family Holdings",
  "amount_usd": 58000, "risk_level": "High", "confidence": 0.86,
  "disposition": "investigate_before_reliance",
  "debate": {
    "challenger": ["related_party", "round_dollar", "outlier", "ghost_vendor", "approval_breach"],
    "defender": { "position": "legitimate_possible", "confidence": 0.25 },
  },
  "verification": { "passed": true, "grounded": "4/4" },
  "citations": ["RP-01 §2", "AP-02 §1", "history:Consulting z=4.1", "vendormaster:absent"],
  "routing": "manual", "escalated_to": "partner",
  "review": { "by": "a.sharma", "action": "reject_pending_evidence", "signed": true,
    "ts": "2026-06-25T11:42:08Z" }
}
```

Audit log entry (immutable)

```
audit_history += {
  "event": "case_closed", "case_id": "CASE-0007",
  "verdict": "High / investigate_before_reliance", "confidence": 0.86,
  "agents": ["supervisor", "evidence", "challenger", "defender", "adjudicator", "verifier"],
  "sources_used": ["RP-01 §2", "AP-02 §1", "vendor_master", "history:Consulting"],
  "human": { "reviewer": "a.sharma(Partner)", "action": "reject_pending_evidence",
    "signature": "sha256:9f2c4b...e71" },
  "timestamp": "2026-06-25T11:42:09Z", "hash_prev": "...", "hash": "..."
}
```